| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1(a) | 4 | **1** | |
| 1(b) | 9 | **1** | |
| 2 | 1, 2, 4, 17, 34, 68 | **2** | **B1** for 5 correct and no extras<br>or 6 correct and 1 extra |
| 3 | $4 \times (6 - 2) + 1 = 17$ | **1** | |
| 4 | $\frac{1}{4}$ or 0.25 | **1** | |
| 5(a) | 576 | **1** | |
| 5(b) | 13 | **1** | |
| 6 | 6.8 | **1** | |
| 7 | 381.42 cao | **1** | |
| 8 | 1098 cao | **2** | **M1** for $\frac{18}{2}(7 \times 18 - 4)$ or better |
| 9 | Correct ruled triangle with construction arcs | **2** | **B1** for correct triangle with incorrect/no arcs or for 2 correct arcs |
| 10 | 12.66 cao | **2** | **B1** for 12.657 to 12.658 or 12.7<br>or **M1** for $\frac{924}{73}$<br>If 0 scored, **SC1** for *their* 12.657… seen and correctly rounded to 2 dp |
| 11(a) | Tangent | **1** | |
| 11(b) | Chord | **1** | |
| 11(c) | 3.72 or 3.720 to 3.721… | **2** | **M1** for $[r^2 =]\ \frac{43.5}{\pi}$ or better |
| 11(d) | 201 or 201.0 to 201.1 | **3** | **B2** for 20.1 or 20.10 to 20.11 [cm] or $64\pi$<br>or **M1** for $6.4 \times \pi$ oe<br>or **B1** for *their* 20.1 [cm] seen and correctly converted to mm |
| 12(a) | 34 | **1** | |

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| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 12(b) | 24 66 | **2** | **B1** for 1 correct |
| 13(a) | 18 | **1** | |
| 13(b)(i) | 1 [hour] 20 [mins] | **2** | **M1** for $\frac{20}{15}$ or 1.333… [hours] |
| 13(b)(ii) | Line from (12 00, 30) to Town D passing through (13 00, 45) | **1** | |
| 13(c) | 72 | **1** | |
| 13(d) | 11.5 or 11.53 to 11.54 | **3** | **B1**FT for 4.333… or 4 hours 20 min<br><br>**M1** for $\frac{50}{3 + \textit{their}\ \textbf{(b)(i)}\ \text{in hours}}$ oe |
| 13(e)(i) | Line from (11 00, 30) to (13 42, 0) | **2** | **B1** for either (11 00, 30) or (13 42, 0) correctly plotted |
| 13(e)(ii) | 27 | **1** | **FT** *their* line in **(e)(i)** |
| 14(a)(i) | 120 | **2** | **M1** for $6 \times 4 \times 5$ oe |
| 14(a)(ii) | $2 \times (6 \times 4 + 4 \times 5 + 6 \times 5)$ leading to 148 | **M2** | **M1** for $6 \times 4$ or $4 \times 5$ or $6 \times 5$ oe soi |
| 14(a)(iii) | 60 | **2** | **M1** for $4 \times 6 + 4 \times 5 + 4 \times 4$ oe |
| 14(b) | $6x + 8y - 2$ final answer | **3** | **B2** for 2 correct terms in final answer or for $6x + 8y - 2$ seen and then spoilt<br><br>or **B1** for either $6x$ or $8y$ or $-2$ in final answer |
| 15(a) | 3.75 or 3.752 to 3.753... | **2** | **M1** for $[r^2 =]\ \frac{177}{4\pi}$ or better |
| 15(b) | 221 or 220.8 to 221.5 | **2** | **FT** *their* radius<br>**M1** for $\frac{4 \times \pi \times (\textit{their}\ \text{radius})^3}{3}$ oe |
| 16(a) | 73 500<br>157 500 | **3** | **B2** for either correct<br>or **M1** for 8 parts = 84 000 soi |
| 16(b)(i) | 6720 | **1** | |
| 16(b)(ii) | 63 | **2** | **M1** for $\frac{56000 - \textit{their}\ 6720 - 14000}{56000}$ [× 100] oe |

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| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 16(b)(iii) | 15 400 or 15 390 or 15 393 or 15 393.[16…] | **2** | **M1** for $14\,000\left(1+\frac{2.4}{100}\right)^4$ oe |
| 17(a) | 144 | **1** | |
| 17(b) | 24 | **2** | **M1** for 2, 2, 2, 3, 5 or better<br>or a correct factor tree or list/table for 120 |
| 17(c) | 720 | **1** | |
| 18(a) | [$a$ =] 13<br>[$b$ =] 19<br>[$c$ =] 25 | **2** | **B1** for 2 terms correct or for 2 terms with a difference of 6 |
| 18(b)(i) | $7n - 3$ oe final answer | **2** | **B1** for $7n + j$ or $kn - 3$ $k \neq 0$ as final answer or for $7n - 3$ oe seen then then further incorrect working |
| 18(b)(ii) | *their* **(b)(i)** = 361 oe | **M1** | |
| | Correct solution leading to 52 | **A1** | |
| 19 | 41.25 | **3** | **M2** for $\frac{12 \times 43.25 + 16 \times 39.75}{28}$ oe<br>or **M1** for $12 \times 43.25$ or $16 \times 39.75$ oe |
| 20 | 1.5 oe | **2** | **B1** for figs 15<br>or **M1** for $\frac{5.4 \times 1000}{60 \times 60}$ oe |
| 21(a) | Angle in a semicircle is 90 | **1** | |
| 21(b) | 7.31 or 7.310 to 7.311 | **3** | **M2** for $\frac{6.2}{\cos 32}$ oe<br>or **M1** for $\frac{6.2}{PQ}$ oe |
| 22 | 5.4[0] or 5.395… | **3** | **M2** for $5.6^2 - 1.5^2$ oe<br>or **M1** for $5.6^2 = [\ldots]^2 + 1.5^2$ oe |

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